Research Statement

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In introductory courses, we usually say that Economics studies the distribution of limited resources with fungible uses to satisfy unlimited needs — I would add, most of the time under conditions of uncertainty. Uncertainty and risk are central to modern economics: the entire expected-utility framework, insurance, and information economics are built around them. Information is key to decisions under uncertainty, and how people use it — or fail to use it — is crucial to understanding behavior and designing better policy. My research sits at the intersection of experimental and behavioral economics, studying how people use information when making decisions and forming beliefs under uncertainty.

When exploration/exploitation decisions do not track changes in information. In my job market paper, I design an incentivized experiment to study sequential experimentation in the canonical one-armed bandit problem (Robbins, 1952; Gittins and Jones, 1974): repeated choice between two arms, one with a known payoff distribution (the familiar arm) and one with an unknown payoff distribution (the unknown arm). Outcomes from the unknown arm are informative about its underlying quality, while outcomes from the familiar arm are not. I test one of the model's central predictions: that agents only reconsider an arm after a change in the information available.

This prediction systematically fails. About half of subjects return to the unknown arm after switching away from it, even though no new information can justify the return, and over-switching persists after uncertainty is resolved: once the unknown arm pays and its quality is known, 39% of subjects still switch away from it at least once. Switching also responds to outcomes that carry no information: a second reward within a visit to an arm lowers the probability of switching away by about 10 percentage points. These deviations distort exploration and are costly: over-switchers explore bad unknown arms about 40% more and earn 20 to 25% less. To probe the mechanism, I use two treatments that limit how switching can respond to realized outcomes: choosing an arm together with the number of rounds to draw from it, and choosing an automatic rule before observing any outcomes. Each

moves about half of the over-switchers to single switching, consistent with noisy execution of a correct plan, while about a quarter keep over-switching under both, consistent with an incorrect model of the problem or non-standard preferences. Subjects consistent with each mechanism also differ in how they switch when choosing freely: those closer to noisy execution switch less often, are less likely to switch right after a failure, and are better predicted by a single-switching plan with occasional slips.

These results motivate two ideas that are currently work in progress. The first one is part of **extensions to strategic experimentation**, in which two subjects face a common unknown arm — its true quality is the same for both — and observe each other's choices and outcomes; I test whether the familiar arm's payoff variability (deterministic vs. stochastic) shapes the free-riding under-exploration predicted in strategic experimentation (Keller, Rady, and Cripps, 2005). The second project is part of **extensions to biased beliefs about random sequences.** I test whether gambler's-fallacy and hot-hand-type distortions in forecasting a known random process — well documented in belief elicitation (Benjamin, 2019) — carry over to a consequential, incentivized choice between lotteries, or whether such distortions are confined to forecasting within a single lottery. Together, this agenda asks when and why people use information—or respond to inputs that are not informative—when making decisions and forming beliefs under uncertainty.

Does Law Have an Informative Function? Across these projects, I study how people use information generated by their own choices and information provided by external institutions when making decisions and forming beliefs. An application of information and uncertainty, directly relevant to policymakers, is how people use legal information to update their beliefs about topics on which they are otherwise poorly informed. In joint work with José Antonio Márquez, Fernando Briseño, and Víctor Pérez, "Legal Information as an Input to Health Risk Assessments: Evidence from E-Cigarettes in Latin America," we test one of the implications of the informative-function-of-law hypothesis of Dharmapala and McAdams (2003) — that people treat the law as a signal about an underlying state, here the health risk of e-cigarettes — using a randomized information experiment fielded across Mexico, Argentina, and Chile, three countries with substantially different and recently defined regulatory regimes. Randomly informing respondents of their country's actual legal status lets us test the causal effect of legal information on perceived health risk and how it varies with baseline beliefs and trust in government.

While we are still completing data collection, preliminary results point toward no significant treatment effect on any risk measure (relative risk, relative addiction, and absolute

risk relative to cigarettes). The clearer differences are across countries: perceptions are more negative in countries where e-cigarettes are banned, even among respondents who report being unfamiliar with the regulation or hold incorrect beliefs about it.

Looking ahead, I want to keep studying the role of information — and of non-informative outcomes — in sequential decision-making and exploration. One direction I'm particularly interested in is how the exploration-exploitation trade-off changes when an agent's own ability influences outcomes ("performance-based bandit"), or, more generally, how people behave in contextual bandits, where the payoff of each arm depends on an observable context that varies from round to round. In addition, going beyond the legal-information project to the broader domain of political beliefs and information, I'm interested in how people update beliefs on contested topics when given information from authoritative sources.

References

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